

Appendix

A. Appendix to Chapter 2

A.1 Survey questions relevant to study (in Dutch)

1. Wat is uw geboortejaar? Gelieve een jaartal op te geven tussen 1901 en 1960.

.....

2. Hoe identificeert u zichzelf?

- ☐ Vrouw
- ☐ Man
- ☐ Anders
- ☐ Zeg ik liever niet

3. Hoe verplaatst u zich het vaakst? Gelieve één antwoordoptie aan te duiden.

- ☐ Te voet
- ☐ Fiets zonder elektrische ondersteuning (bv. stadsfiets, plooi-fiets, koersfiets...)
- ☐ Fiets met elektrische ondersteuning (bv. elektrische fiets, speed pedelec...)
- ☐ Openbaar vervoer (bus, tram, trein)
- ☐ Persoonlijke wagen als chauffeur
- ☐ Persoonlijke wagen als passagier (bv. bij partner of familielid)
- ☐ Taxi
- ☐ Andere, namelijk (gelieve aan te vullen):

4. In welke buurt van Gent woont u?

- | | |
|---|---|
| ☐ Binnenstad | ☐ Muide – Meulestede – Afrikalaan |
| ☐ Brugse Poort - Rooigem | ☐ Nieuw Gent – UZ |
| ☐ Bloemekenswijk | ☐ Oostakker |
| ☐ Dampoort | ☐ Oud Gentbrugge |
| ☐ Drongen | ☐ Rabot – Blaisantvest |
| ☐ Elisabethbegijnhof – Prinsenhof –
Papegaai – Sint-Michiels | ☐ Sint-Amandsberg |
| ☐ Gentse Kanaaldorpen en -zone | ☐ Sint-Denijs-Westrem – Afsnee |
| ☐ Gentbrugge | ☐ Sluizeken – Tolhuis – Ham |
| ☐ Ledeberg | ☐ Stationsbuurt-Noord |
| ☐ Macharius – Heirnis | ☐ Stationsbuurt-Zuid |
| ☐ Mariakerke | ☐ Watersportbaan – Ekkerghem |
| ☐ Moscou – Vogelhoek | ☐ Wondelgem |
| | ☐ Zwijnaarde |

5. In welke mate gaat u akkoord met volgende stellingen over fietsen in uw buurt? Gelieve voor elke stelling aan te duiden in welke mate u hiermee akkoord gaat.

	Helemaal niet akkoord	Niet akkoord	Neutraal	Akkoord	Helemaal akkoord
Ik woon in een fietsvriendelijke buurt.	☐	☐	☐	☐	☐
Ik vind het aangenaam om in mijn buurt te fietsen.	☐	☐	☐	☐	☐
De fietspaden in mijn buurt zijn van goede kwaliteit.	☐	☐	☐	☐	☐
De fietsinfrastructuur in mijn buurt is goed onderhouden.	☐	☐	☐	☐	☐
Er is voldoende fietsinfrastructuur in mijn buurt (fietspaden, fietssuggestiestroken, fietsstraten, fietsonderdoorgangen en fietsbruggen).	☐	☐	☐	☐	☐
De straten en pleinen zijn in goede staat in mijn buurt.	☐	☐	☐	☐	☐
Het verkeer in mijn buurt is heel druk waardoor ik mij onveilig voel als ik fiets.	☐	☐	☐	☐	☐
Ik voel me als fietser veilig in het verkeer in mijn buurt.	☐	☐	☐	☐	☐

6. In welke mate gaat u akkoord met onderstaande stellingen over wandelen in uw buurt? Gelieve voor elke stelling aan te duiden in elke mate u hiermee akkoord gaat.

	Helemaal niet akkoord	Niet akkoord	Neutraal	Akkoord	Helemaal akkoord
Ik woon in een wandelvriendelijke buurt.	☐	☐	☐	☐	☐
In mijn buurt is het gemakkelijk om rond te wandelen met een rollator of rolstoel.	☐	☐	☐	☐	☐
De voetpaden in mijn buurt zijn goed onderhouden.	☐	☐	☐	☐	☐
In mijn buurt is er weinig hinder op het voetpad (paaltjes, fietsen tegen de gevels,...).	☐	☐	☐	☐	☐
In mijn buurt zijn er voldoende zitbanken.	☐	☐	☐	☐	☐
Voetgangers kunnen in mijn buurt vlot oversteken (zebrapaden, vlakke oversteekplaatsen, verlaagde stoepranden).	☐	☐	☐	☐	☐
Er is voldoende groen (bomen, parken, bloemperkjes,...) in mijn buurt.	☐	☐	☐	☐	☐
Openbare plekken in mijn buurt (zoals parken en pleinen) zijn gezellige ontmoetingsplaatsen.	☐	☐	☐	☐	☐
Er wordt voldoende verlichting voorzien op openbare plaatsen als het donker is (bv. op straat, aan bushaltes, oversteekplaatsen).	☐	☐	☐	☐	☐
Ik voel me als voetganger veilig in het verkeer in mijn buurt.	☐	☐	☐	☐	☐

7. In welke mate gaat u akkoord met volgende stellingen rond de wagen in de buurt? Gelieve voor elke stelling aan te duiden in welke mate u hiermee akkoord gaat.

	Helemaal niet akkoord	Niet akkoord	Neutraal	Akkoord	Helemaal akkoord
Mijn buurt is vlot bereikbaar met de wagen.	☐	☐	☐	☐	☐
In mijn buurt is er veel verkeer.	☐	☐	☐	☐	☐
In mijn buurt is er veel vrachtverkeer.	☐	☐	☐	☐	☐
Er is veel hinder (geluidsoverlast, geurhinder, trillingen) veroorzaakt door het verkeer in mijn buurt.	☐	☐	☐	☐	☐
Er rijden voldoende taxi's in mijn buurt.	☐	☐	☐	☐	☐
In mijn straat kunnen bewoners gemakkelijk een parkeerplaats vinden.	☐	☐	☐	☐	☐
Mijn bezoekers kunnen in mijn straat gemakkelijk een parkeerplaats vinden.	☐	☐	☐	☐	☐
Ik voel me als autogebruiker veilig in het verkeer in mijn buurt. (Enkel in te vullen als u verplaatsingen maakt met de auto.)	☐	☐	☐	☐	☐
Er vinden weinig verkeersongevallen plaats in mijn buurt.	☐	☐	☐	☐	☐

8. In welke mate gaat u akkoord met volgende stellingen rond het openbaar vervoer in uw buurt? Gelieve voor elke stelling aan te duiden in welke mate u hiermee akkoord gaat.

	Helemaal niet akkoord	Niet akkoord	Neutraal	Akkoord	Helemaal akkoord
Ik ben tevreden met het OV in mijn buurt.	☐	☐	☐	☐	☐
In mijn buurt zijn er voldoende bus-en/of tramlijnen.	☐	☐	☐	☐	☐
In mijn buurt rijden bussen/trams voldoende frequent.	☐	☐	☐	☐	☐
De bus/tramhaltes in mijn buurt zijn goed uitgerust (schuilhokje, vuilbak, zitbank,...).	☐	☐	☐	☐	☐
De bus/tramhaltes in mijn buurt zijn voldoende toegankelijk voor iedereen (bv. met een rolstoel, kinderwagen,...).	☐	☐	☐	☐	☐
Er zijn voldoende zitplaatsen beschikbaar op bussen en trams.	☐	☐	☐	☐	☐
Ik voel me veilig op het OV.	☐	☐	☐	☐	☐
Het is gemakkelijk voor ouderen om informatie op te zoeken rond het OV.	☐	☐	☐	☐	☐

A.2 ANOVA mean-comparison tests – results

ANOVA mean-comparison tests based on the calculated factor scores were performed to explore how the living environment of older adults relates to the main factors. Post-hoc tests were performed for the living environment variables. First, to decide which post-hoc tests were the most appropriate, Levene's tests were completed to determine whether the groups had homogeneity of variance. Homogeneity of variance means that the variance among the compared groups should be approximately equal. If the p-value is greater than 0.05, this suggests that the variances are equal across groups, indicating that the assumption of homogeneity has not been violated. Tukey's post-hoc tests were performed for those with assumed homogeneity of variance from non-significant Levene's tests. Dunnett's C post-hoc tests were performed for those with assumed unequal variance from significant Levene's tests.

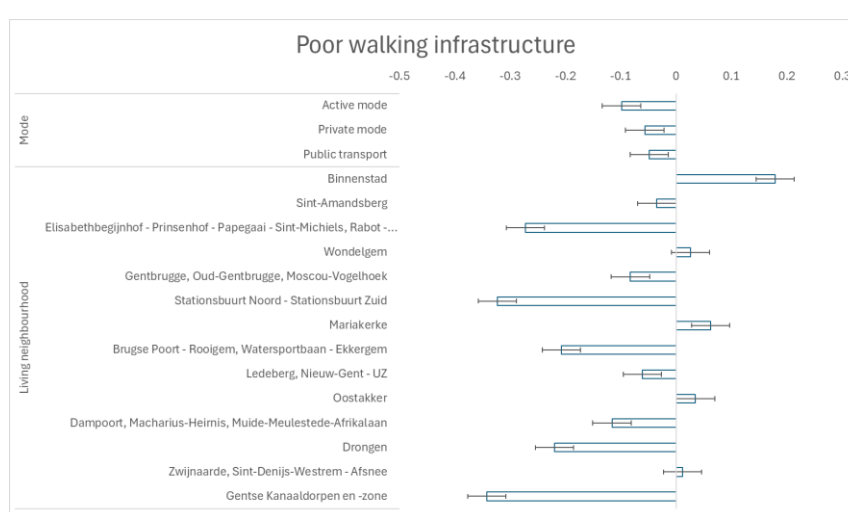


Figure A.1: ANOVA results for "Poor walking infrastructure"

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene's test "Mode" p -value = 0.798; Levene's test "Living neighbourhood" p -value = 0.101

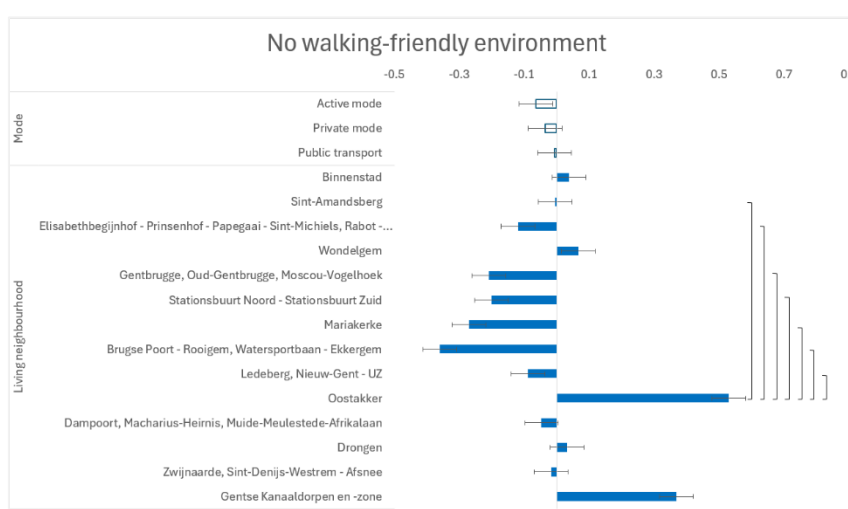


Figure A.2: ANOVA results for "No walking-friendly environment"

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene's test "Mode" p -value = 0.512; Levene's test "Living neighbourhood" p -value = 0.133

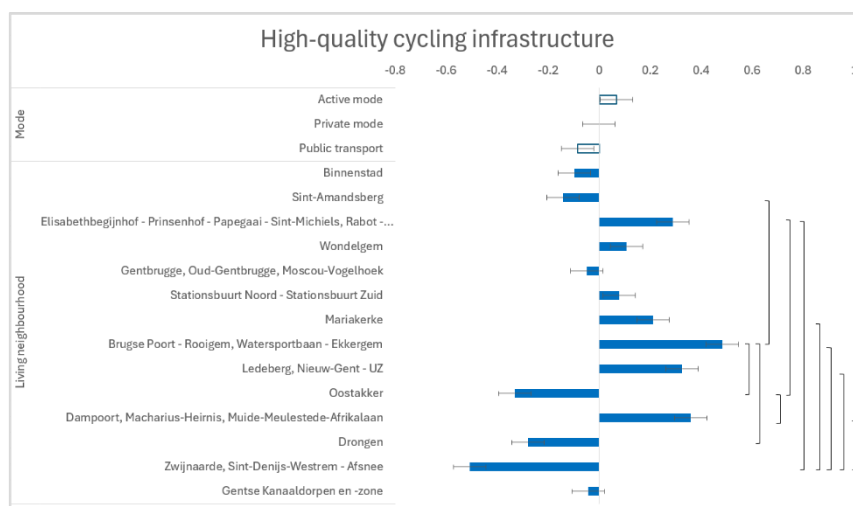


Figure A.3: ANOVA results for “High-quality cycling infrastructure”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.099; Levene’s test “Living neighbourhood” p -value = 0.379

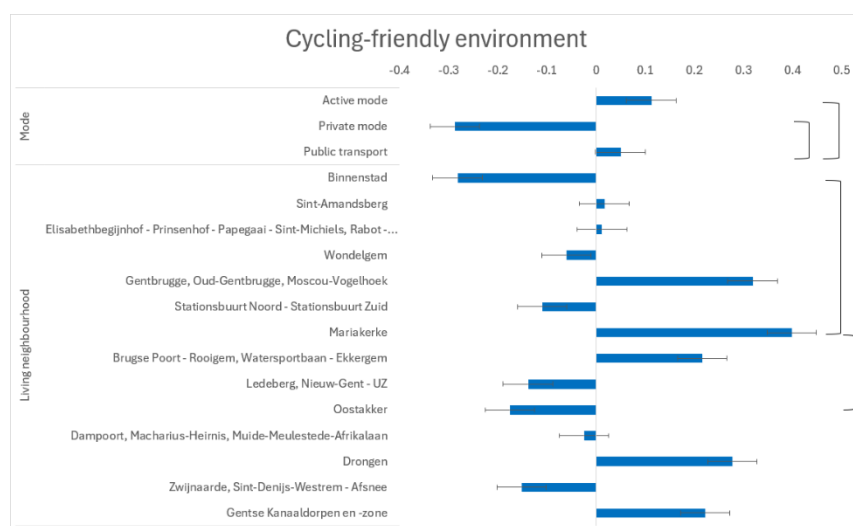


Figure A.4: ANOVA results for “Cycling-friendly environment”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.988; Levene’s test “Living neighbourhood” p -value = 0.386

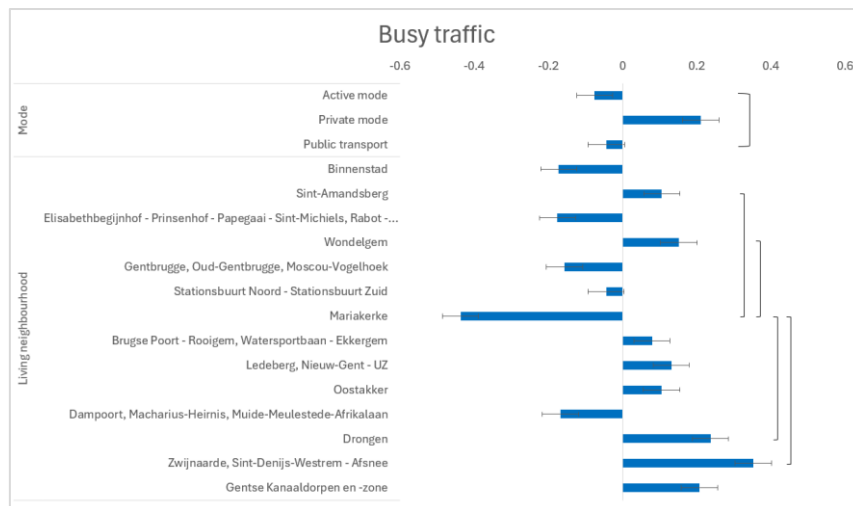


Figure A.5: ANOVA results for “Busy traffic”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.123; Levene’s test “Living neighbourhood” p -value = 0.473

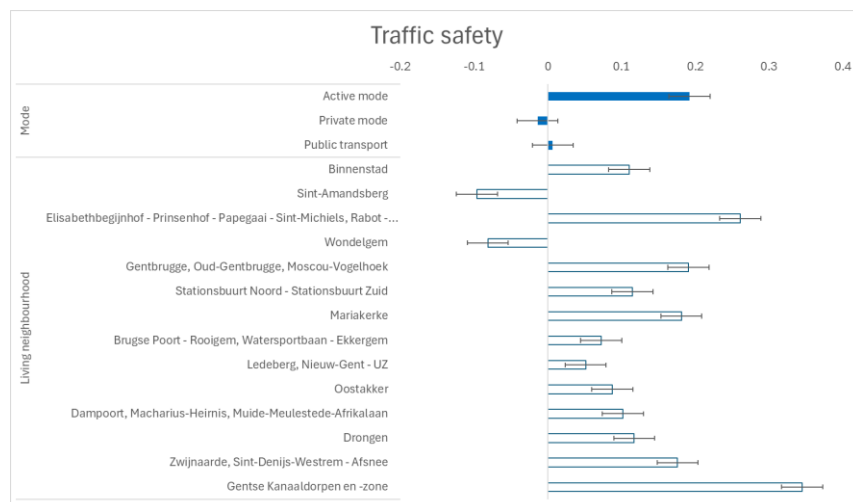


Figure A.6: ANOVA results for “Traffic safety”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.012; Levene’s test “Living neighbourhood” p -value = 0.012

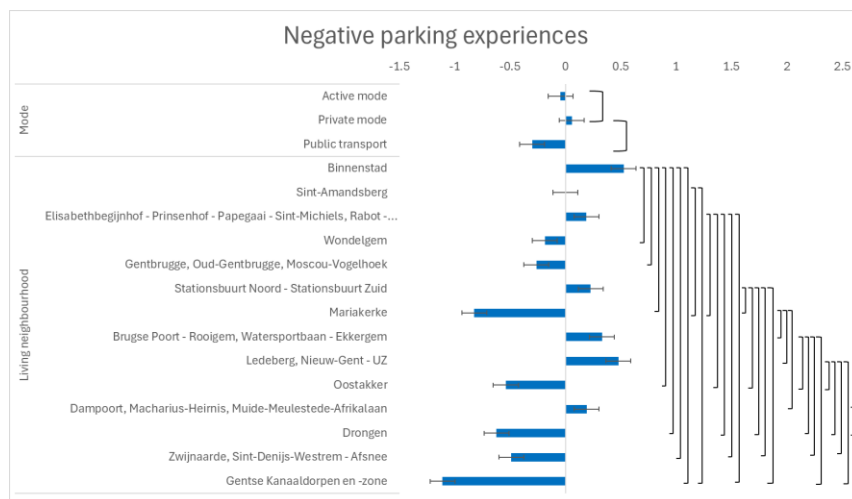


Figure A.7: ANOVA results for “Negative parking experiences”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.446; Levene’s test “Living neighbourhood” p -value = 0.002

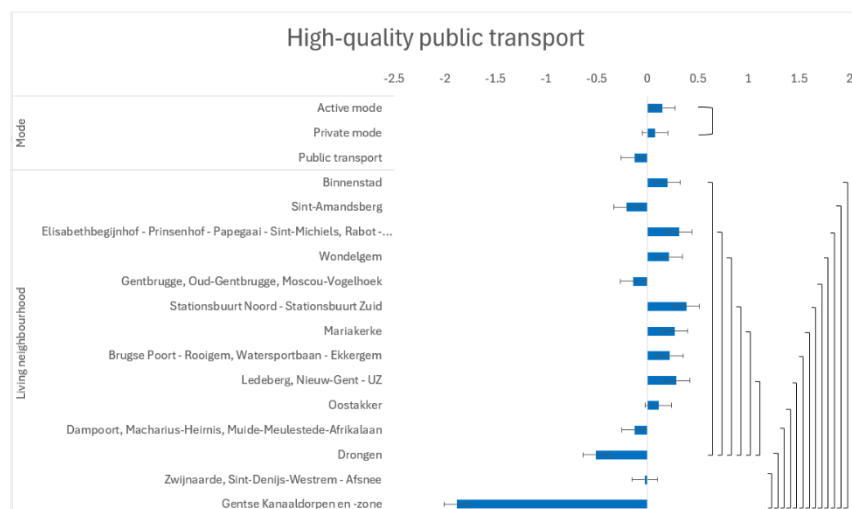


Figure A.8: ANOVA results for “High-quality public transport”

Note: Blue bars: $p < 0.05$; White bars: no significant differences; Brackets show significant differences between groups. Levene’s test “Mode” p -value = 0.085; Levene’s test “Living neighbourhood” p -value = 0.004

A.3 Spatial distribution of objective environment measures

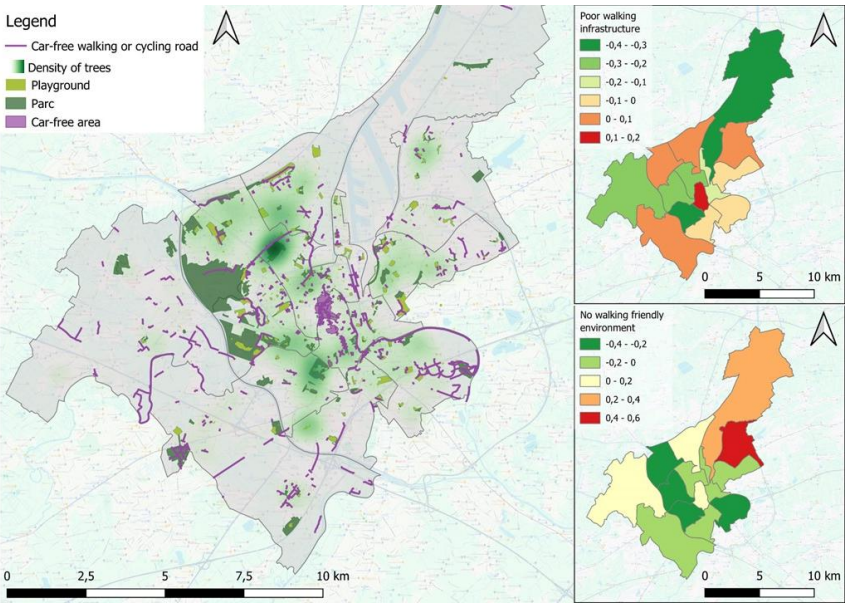


Figure A.9: Objective built environment measures related to walking

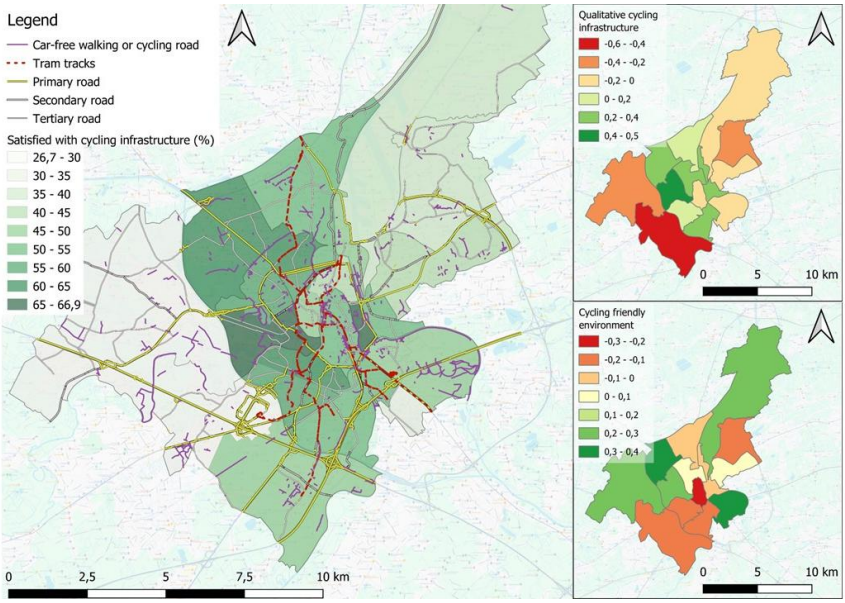


Figure A.10: Objective built environment measures related to cycling

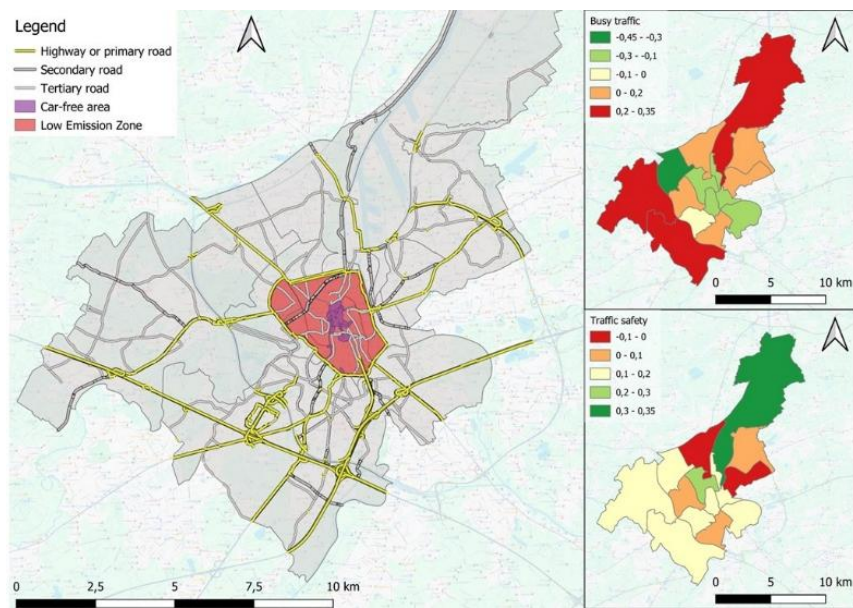


Figure A.11: Objective built environment measures related to traffic

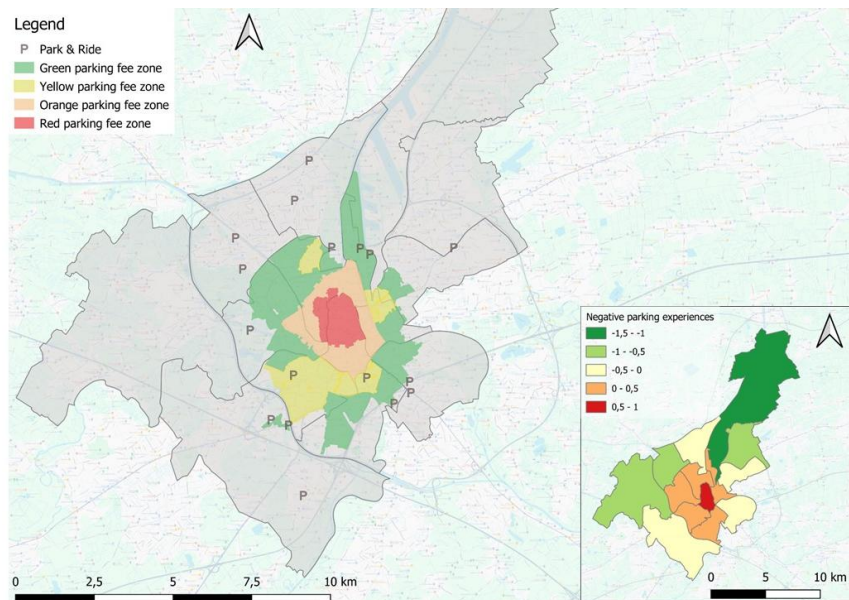


Figure A.12: Objective built environment measures related to parking

A.4 Within- and between-neighbourhood associations

Table A.1: Within- and between-neighbourhood associations

Variable	Walking vs car	Cycling vs car	Public transport vs car
Individual-level covariates			
Age	0.77 (0.52–1.12)	0.73 (0.56–0.96)	1.25 (0.90–1.76)
Gender: male	0.87 (0.47–1.60)	0.95 (0.61–1.48)	0.42 (0.22–0.78)
Educational level	0.98 (0.67–1.44)	1.25 (0.95–1.65)	0.75 (0.54–1.04)
Household size	1.31 (1.01–1.75)	1.00 (0.76–1.32)	0.99 (0.74–1.33)
EQ-5D index	1.59 (1.05–2.49)	1.29 (0.95–1.75)	0.99 (0.70–1.38)
Household income	0.85 (0.60–1.19)	0.95 (0.76–1.19)	0.90 (0.66–1.21)
Within-neighbourhood perceived environment			
High-quality cycling infrastructure	1.65 (1.01–2.71)	0.80 (0.54–1.18)	1.47 (0.91–2.34)
Cycling-friendly environment	0.56 (0.34–0.93)	1.36 (0.91–2.05)	0.56 (0.34–0.90)
Positive parking experiences	0.91 (0.66–1.25)	1.01 (0.81–1.27)	1.18 (0.87–1.57)
Between-neighbourhood perceived environment			
High-quality cycling infrastructure	1.68 (0.81–3.42)	1.09 (0.67–1.78)	1.31 (0.67–2.50)
Cycling-friendly environment	0.81 (0.40–1.65)	1.22 (0.78–1.96)	1.17 (0.62–2.25)
Positive parking experiences	0.37 (0.23–0.60)	0.59 (0.43–0.80)	0.41 (0.27–0.62)

Note: Values are odds ratios (OR) with 95% Bayesian credible intervals (CrI). Bold values indicate that the original, unrounded 95% CrI excludes OR = 1. Within-neighbourhood effects represent individual deviations from the neighbourhood mean; between-neighbourhood effects represent neighbourhood mean perceptions. Private car is the reference outcome category.

A.5 ICLV benchmark – sensitivity analysis

Table A.2: Associations between latent constructs and transport mode choice

Construct	Walking vs car	Cycling vs car	Public transport vs car
Cycling-friendly environment	0.52 (0.31–0.85)	1.45 (0.99–2.11)	0.56 (0.35–0.90)
High-quality cycling infrastructure	2.06 (1.26–3.36)	0.94 (0.68–1.30)	1.72 (1.15–2.59)
Positive parking experiences	0.52 (0.37–0.72)	0.78 (0.62–0.98)	0.73 (0.54–0.98)

Note: Values are odds ratios (OR) with 95% confidence intervals (CI). Bold values indicate that the original, unrounded 95% CI excludes OR = 1. Private car is the reference outcome category.

Table A.3: Measurement model loadings

Construct	Indicator	Loading	Robust SE	95% CI
High-quality cycling infrastructure	Maintenance of cycling infrastructure	5.37	0.69	4.02–6.71
High-quality cycling infrastructure	Quality of cycling lanes	4.52	0.49	3.57–5.48
High-quality cycling infrastructure	Maintenance of streets and squares	1.80	0.16	1.48–2.12
High-quality cycling infrastructure	Sufficient cycling infrastructure	3.00	0.26	2.50–3.50
Cycling-friendly environment	Busy traffic causing unsafe cycling experiences	-1.55	0.18	-1.90–1.20
Cycling-friendly environment	Enjoyable cycling	3.92	0.49	2.95–4.88
Cycling-friendly environment	Cycling-friendly neighbourhood	3.62	0.42	2.79–4.44
Cycling-friendly environment	Feeling safe in traffic as cyclist	2.24	0.22	1.81–2.67
Positive parking experiences	Accessible by car	0.86	0.10	0.66–1.07
Positive parking experiences	Easy parking for residents	4.99	0.63	3.76–6.22
Positive parking experiences	Easy parking for visitors	10.38	3.15	4.21–16.56

Note: SE = standard error; CI = confidence interval. Robust standard errors are reported. The measurement model used ordinal indicators with five response categories.

B. Appendix to Chapter 3

B.1 Clusters of neighbourhoods based on the presence of transport infrastructure

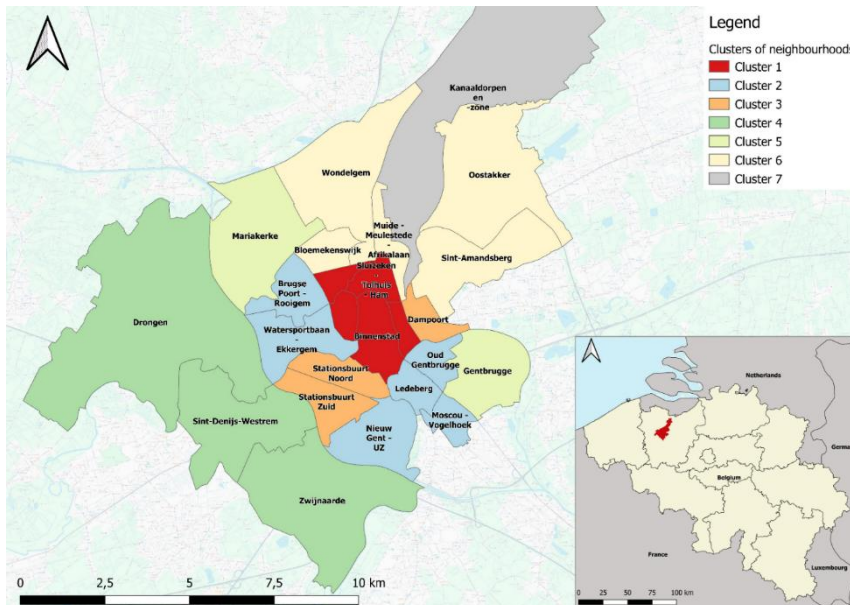


Figure B.1: Clusters of neighbourhoods based on the presence of transport infrastructure

Compared to Figure 3.1 (Clusters of neighbourhoods), this map contains two major differences. First, the neighbourhoods of Mariakerke and Gentbrugge have been separated from the clusters adjacent to the city centre, mainly due to the presence of cycling infrastructure within the two nature reserves located in these areas. Second, the neighbourhoods containing Ghent's main railway stations have been grouped into a distinct cluster (Cluster 3), reflecting the strong provision of public transport.

B.2 Differences in coding interpretation and refinement of coding framework

Table B.1: Differences in coding interpretation and refinement of coding framework

Quote	Reviewer 1*	Reviewer 2*	Decision*	Discussion of coding interpretations and refinement of coding framework
More green spaces	3	3, 4	3	'More green spaces' refers to the provision of greenery in public space. If liveability of the neighbourhood is not specifically mentioned, these kinds of quotes fit in the topic of public space.
Keep sidewalks free of weeds, illuminate crossings, reduce car speeds, increase police presence in short, make the environment safer, not just in theory politics but also in reality	1	1, 3	1, 3	Multiple topics discussed in the quote.
Provide better sidewalks and ensure that cyclists follow the same direction as cars. Also, provide parking spaces for residents, as placing trees in the street is not safe	1	1, 3, 4	1, 3, 4	Multiple topics discussed in the quote.
Quieter traffic, meaning car drivers who race through the street	4	1	1, 4	Quote refers both to the liveability of the neighbourhood and the enforcement of traffic rules.
Dangerous cyclists, scooter riders, and poor road conditions are the biggest problems	1	1, 3, 4	1, 3	Multiple topics discussed in the quote.
Better and separated bike lanes	3	1, 3	1, 3	Multiple topics discussed in the quote.
Better sidewalks, better maintenance of greenery, and teaching cyclists to follow traffic rules	3	1, 3	1, 3	Multiple topics discussed in the quote.
Make the intersection at Lousbergskaaai Kasteellaan safe for cyclists	3	1	1	Safety at intersections refers to possible conflicts between modes and overall traffic dynamics.
Fewer detours, fewer traffic jams, more greenery	4	1, 4	1, 4	Multiple topics discussed in the quote.
Speed controls for cars control bikes riding in the opposite direction on bike paths ban loud mopeds and cars increase cleanliness illegal dumping, abandoned bikes, leftover traffic signs after construction, etc	4	1, 3, 4	1, 3, 4	Multiple topics discussed in the quote.
A smooth bike path between the tram tracks	3	1, 3	1, 3	Multiple topics discussed in the quote.
Keep sidewalks in good condition improve boarding options for buses and trams	2	2, 3	2, 3	Multiple topics discussed in the quote.
More green spaces, easily accessible on foot and also maintained	4	3, 4	3, 4	Multiple topics discussed in the quote.
People should be encouraged to throw their waste into designated containers and not on the street, as often happens in my neighbourhood	4	3	1, 3	Both enforcement (due to the need of encouragement) and the maintenance of public space in the neighbourhood are discussed.
More parking throughout Ghent	3	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
Sheltered bus stops, more benches, well maintained sidewalks	3	2, 3	2, 3	Multiple topics discussed in the quote.
Better constructed sidewalks and closer bus tram stops	2	2, 3	2, 3	Multiple topics discussed in the quote.
Better sidewalks and bike paths in my street and less heavy traffic such as trucks	3	1, 3	1, 3	Multiple topics discussed in the quote.

Parking spaces	3	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
bus drivers should be more careful and polite	4	2	2	Bus drivers are part of the public transport services, and not of the social dynamics of the neighbourhood itself.
Urgent repaving of cobblestone paths is needed fewer terraces that currently occupy a large portion of the sidewalks and every tram and bus stop should have proper shelter and a screen displaying the timetable	3	2, 3	2, 3	Multiple topics discussed in the quote.
Definitely enough parking spaces, especially for the disabled, and not a parking tower miles away from the neighbourhood	3	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
Remove the tram from the Korenmarkt in the city centre	3	2	2	Topics related to the addition or removal of public transport (stops) are assigned to 'Public transport accessibility'.
Within the apartment block, I would like to have more activities with other residents I would like better sidewalks in the area	4	3, 4	3, 4	Multiple topics discussed in the quote.
More fines, good sidewalks and cycling Increase cyclists and scooter riders' awareness of traffic rules, and the same for illegal dumpers	4	1, 3	1, 3	The mentioning of fines and enforcement of traffic rules is related to urban traffic dynamics. The provision of sidewalks refers to the public space.
Improve sidewalks and make public transport safer for less mobile seniors, especially when getting off	2	2, 3	2, 3	Multiple topics discussed in the quote.
Better enforcement of parking bans in certain streets	4	1	1	The quotes stress the need for enforcement.
Nothing for now I feel good in my home, and I realize I'm lucky in that if I face difficulties, I might look for another home closer to my children or cohabit with someone since I realize that this might be better as I grow older and need more help Friends and I used to joke about plans for a House of Laughter and Joy, and who knows, it might still happen	5	4	4, 5	Both no need for changes and possible future changes are discussed.
Better maintained sidewalks and cleaner bus stops and buses	3	2, 3	2, 3	Multiple topics discussed in the quote.
MORE GREEN SPACES AND ACCESSIBLE BENCHES FOR WHEELCHAIR USERS AS WELL AND PICNIC TABLES	4	3, 4	3, 4	Multiple topics discussed in the quote.
Much better sidewalks, especially to tram line 1 you have to walk on the street and make existing sidewalks flat and accessible for wheelchairs	3	2, 3	2, 3	Multiple topics discussed in the quote.
Maintenance of roads, keeping out cut through traffic, bike and footpaths this is not all in the rural part of my street There are 30 cm deep potholes along the side of the street, and its been like this for over 15 years	3	1, 3	1, 3	Multiple topics discussed in the quote.
Clean streets, remove stinky garbage from sidewalks, and fine bike terrorists	3	1, 3	1, 3	Multiple topics discussed in the quote.
Access to public transport distance Enforce rules for cyclists running red lights, giving right of way, riding on the right Ban e scooters Provide parking	- 1 -	1, 2, 4	1, 2, 4	Multiple topics discussed in the quote.

More parking spaces for cars and spaces for the disabled	2	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
More parking space for residents, fewer shared cars	1	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
Possibly reduce parking pressure, as there are too few parking spaces in the area UZ hospital, many students at the Sterre, social housing without garages, resulting in cars often parked too close to or in front of our garage	1	4	4	Consensus to refer all quotes related to parking issues to the theme of 'Neighbourhood and social dynamics' since this mainly refers to the liveability and accessibility of the neighbourhood.
More benches and better sidewalks Public transport that's easier for elderly people to get on and off, like at the seaside, where its level	3	2, 3	2, 3	Multiple topics discussed in the quote.
Better and safer cycling infrastructure cycling streets, wider bike lanes, better signage at construction sites, speed limits for cars, more greenery	1	1, 3, 4	1, 3, 4	Multiple topics discussed in the quote.
Better sidewalks and bike paths, and have our bus stops in the neighbourhood restored	3	2, 3	2, 3	Multiple topics discussed in the quote.

* 1: Urban traffic dynamics; 2: Public transport accessibility; 3: Public space and maintenance; 4: Neighbourhoods and social dynamics; 5: No changes

B.3 Biterm Topic Models of neighbourhood clusters

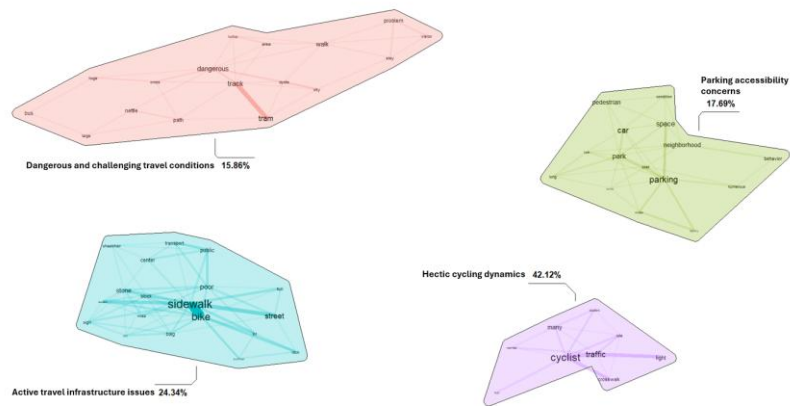


Figure B.2: Biterm topic model of cluster 1 - city centre neighbourhoods

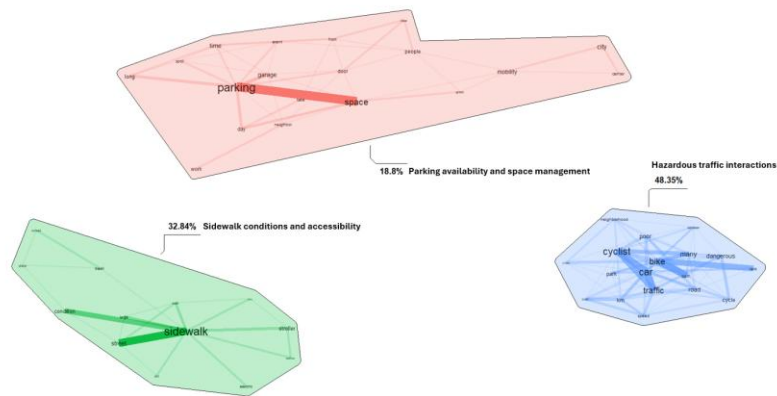


Figure B.3: Biterm topic model of cluster 2 – neighbourhoods adjacent to city centre

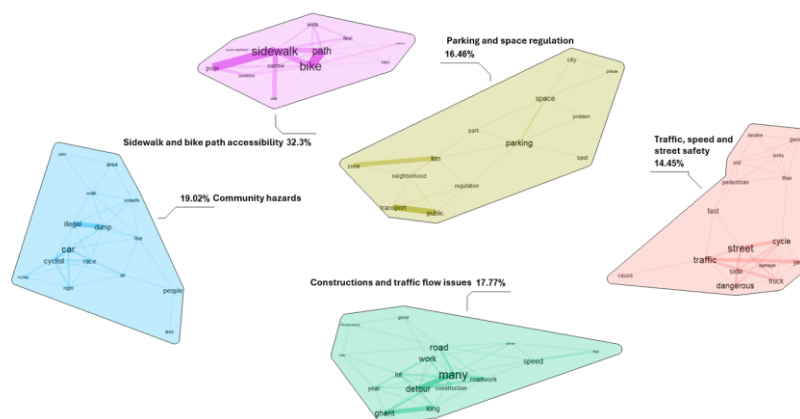


Figure B.4: Biterm topic model of cluster 3 – Suburban residential neighbourhoods

B.4 Biterm Topic Models of gender and age groups

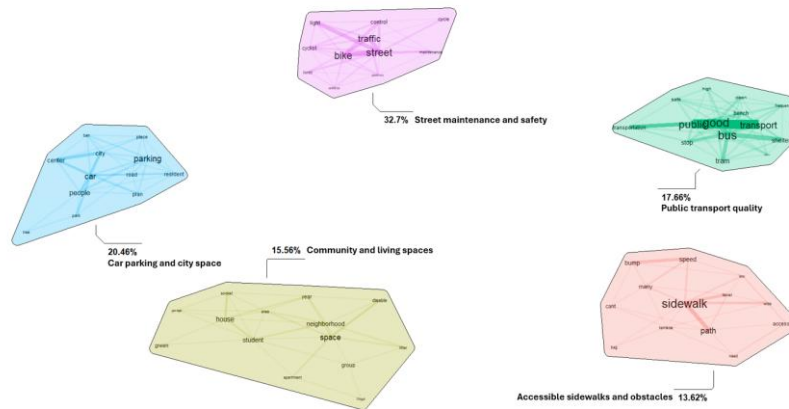


Figure B.6: Biterm topic model of gender – women

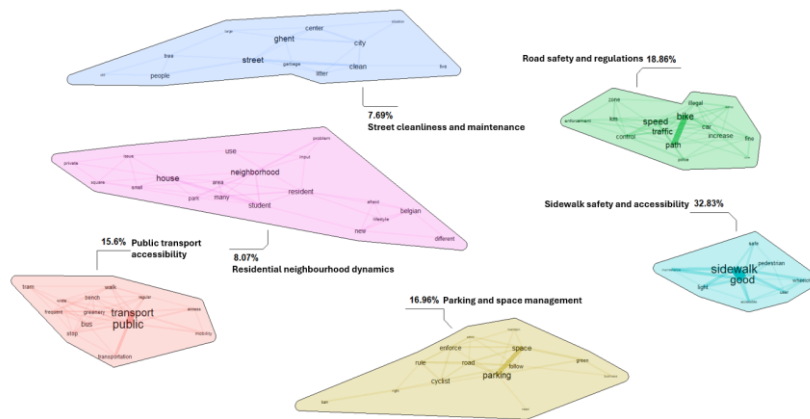


Figure B.7: Biterm topic model of gender – men

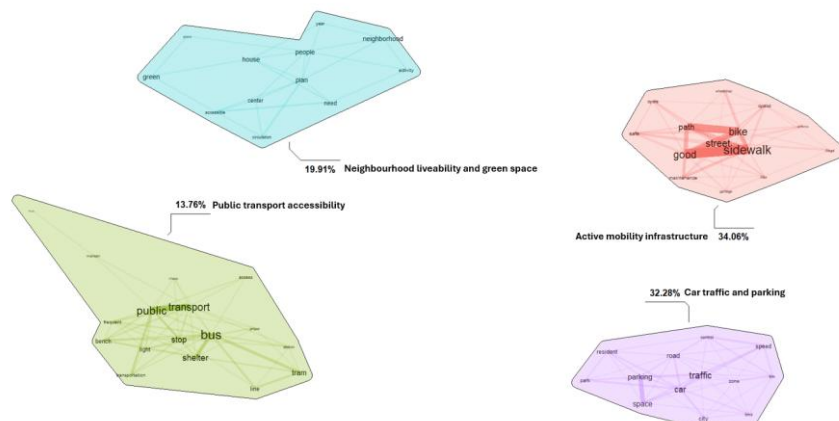


Figure B.8: Biterm topic model of age groups – younger than the median age of 71 (1953)

B.5 UMass coherence scores BTM and LDA

Table B.2: UMass coherence scores BTM and LDA

	Coherence score BTM	Coherence score LDA
Topic 1	-1.981659	-3.090716
Topic 2	-2.463689	-3.079150
Topic 3	-2.327179	-2.437123
Topic 4	-2.055153	-3.117081
AVERAGE	-2.20692	-2.931018

B.6 Themes and quotes – Urban traffic dynamics

Table B.3: Urban traffic dynamics (N = 266 – 20.18%)

Theme	Number of quotes	Subtheme	Number of quotes	Example quotes
Need for control, sanctions, enforcement, and education	86 (32.33%)			• "Make sure that cyclists, cargo bikers, scooter riders and ebikers follow the traffic rules. Act consistently and decisively against reckless road users by giving them heavy fines." • "Enforce stricter measures against violations of public spaces, including misbehaviour by cyclists." • "More police on the streets. Strictly fining offenders. Every bike should have a license plate, so they are no longer anonymous and must follow the same rules as other road users."
Conflicts between different modes resulting in unsafe travelling conditions	64 (24.06%)			• "I would try to better structure the space for all road users. In my neighbourhood, there is sometimes real chaos due to the large number of road users. You can't lose focus for even a minute and have to be constantly alert while moving around." • "Better spatial planning to reduce driveway intersections on bike paths." • "People walk in the middle of the street in the city centre instead of on the sidewalks, making cycling difficult." • "Cyclists have no respect for pedestrians, so clear separation is needed, also physically between them."
Speed limits	20 (7.52%)	Speed limit is too high (90.00%)	18 (90.00%)	• "Cars should drive slower." • "Put more restrictions on car traffic, especially speed." • "Reduce car traffic and, above all, slow it down."
		Speed limit is too low (10.00%)	2 (10.00%)	• "Abolish the 30 km/h zone on main roads and only apply it near schools, hospitals etc."
Restriction on certain transport modes	20 (7.52%)	Restriction on cut through car traffic (55.00%)	11 (55.00%)	• "Keep traffic out of the village centre to reduce cut through traffic." • "Car traffic in the area should be reduced due to regular congestion."
		Restriction on trucks (25.00%)	5 (25.00%)	• "No trucks in residential areas. It is often used as a shortcut." • "Adjust truck routes."
		Restriction on (e-)scooters, speed pedelecs, or cyclists (20.00%)	4 (20.00%)	• "Ban electric scooters in the city." • "Less busy bicycle traffic."

Mismanagement of the city	18 (6.77%)			• "Less interference from the city council. It has a negative effect on the neighbourhood." • "To start, NO unnecessary spending on e.g., useless bicycle bridges. A dept of over a billion is enough for the next generations who aren't aware of it yet." • "Too many changes are being made that, in my opinion, are not always thought through and are unnecessary. Too much attention is given to a certain target group." • "The Ghent government doesn't listen anyway."
Adaptations to or abolishment of mobility plans / circulation plan / LEZ	18 (6.77%)			• "Abolish the circulation plan and do not allow the closure of Gentbrugge bridge." • "Adjust and reduce the mobility plan." • "Make the circulation plan more efficient so unnecessary detours are not needed."
Other traffic regulations	14 (5.26%)			• "Make the city more accessible to motorists so that commerce can flourish again." • "Motorcycles should be quieter." • "Make driving more expensive."
Detours, road closures, and congestion	11 (4.14%)			• "Better coordination over different construction sites and better road signage." • "Stop all road closures for cars. Soon, my street will have its direction changed 3 times, meaning I will have to drive 900 meters instead of 200." • "Smoother traffic in Ghent by removing certain road closures."
Need for traffic calming measures	10 (3.76%)			• "Place flower boxes on the road to slow down traffic." • "A few speed bumps in our newly constructed street. There are many speed demons." • "Use parked cars as traffic calming measures."
Need for more car-oriented planning (focusing on older adults)	5 (1.88%)			• "Consider people who need their car to get around. I think there is too much priority for cyclists." • "Don't forget that some people need to use a car to get outside. Not everyone is mobile enough to cycle or walk." • "Ensure that older people can move around in the city centre by car, as not everyone is mobile or able to bike."

B.7 Themes and quotes – Public transport accessibility

Table B.4: Public transport accessibility (N = 174 – 13.20%)

Theme	Number of quotes	Subtheme	Number of quotes	Example quotes
General quality of public transport	36 (20.69%)			“Public transport needs to be better.” “Improvement of public transport.”
Accessibility to stops	32 (18.39%)			“More regular public transport within walking distance instead of abolishing it.” “Bring public transport closer.”
Frequency	30 (17.24%)			“Make public transport more frequent and available at wider times for train, bus, and tram.” “More trams, buses, and trains in the evening after cultural activities.” “Every 20 minutes a tram or bus on all lines until 1 a.m. seems like an ideal starting point.”
Abolishment of public transport stops	21 (12.07%)			“Have our bus stops in the neighbourhood restored.” “Maintain the current number of bus stops.” “Bring back our buses.”
Sheltered public transport stops	15 (8.62%)			“Sheltered bus stops everywhere and benches here and there along the way.” “Add benches and shelters at all stops.” “Larger bus and tram shelters. Often you are partially in the rain and wind. Many bus and tram shelters are poorly lit, making it feel unsafe to wait for a bus or tram in the evening.”
Getting on/of the bus	12 (6.90%)			“Make bus and tram stops accessible. Currently, you often have to take a leap down or make a high step. It would also be nice for people with strollers.” “More public transport and higher platforms. Sometimes I have to jump off the bus, and it is painful.”
Connections	12 (6.90%)	Connections in general	7 (58.33%)	“Better bus connections so that people don’t have to transfer as much and lose less time.” “Better connections with public transport.”
		Connection to city centre	4 (33.33%)	“Better optimised city transport means, especially for this neighbourhood with many seniors, that a bus should go to the Vrijdagmarkt and/or Korenmarkt.” “Larger supply of public transport towards the centre of Ghent.”
		Connection to train station	1 (8.33%)	“More direct buses to Ghent Sint-Pieters [main train station of Ghent].”

Driving behaviour and safety on the bus	4 (2.30%)			• "And more cautious tram and bus drivers. They are often cowboys." • "Above all, make bus seats safer by providing benches. Some steps are high, and seats should be safer when the bus has to brake suddenly. I always found it odd that seat belts aren't mandatory on buses."
Removal of trams/ tram tracks	3 (1.72%)			• "Remove tram tracks and use electric buses instead." • "No extra tram tracks due to safety concerns."
Pricing of public transport	3 (1.72%)			• "Make public transport cheaper [...] Give us transport like ministers and the royal family have. They earn a lot and still get free transport." • "Cheaper and better public transport for everyone."
Taxis	2 (1.15%)			• "Cheaper taxis that can drive everywhere." • "Make regular taxis cheaper."
Interruptions of public transport	2 (1.15%)			• "Long interruptions in transport options due to constructions of De Lijn e.g., Coupure Rechts." • "The tram line should remain active along the entire line 1, without digging up a street for two years preventing it from passing."
Mismanagement	2 (1.15%)			• "De Lijn – the public transport company – is insufficiently subsidised and has weak policies."

B.8 Themes and quotes – Public space and maintenance

Table B.5: Public space and maintenance (N = 494 – 37.48%)

Theme	Number of quotes	Subtheme	Number of quotes	Example quotes
Maintenance and provision of sidewalks	211 (42.71%)	Maintenance and provision of sidewalks overall	176 (83.41%)	“Sidewalks, no longer having to walk on the street. Replace cobblestones in the village with tiles.” “Safe sidewalks, well lit, fix loose stones and fill potholes.” “Pay more attention to overcrowded sidewalks. Remove obstacles such as temporarily placed traffic signs.” “Daily cleanup of litter and illegal dumping, maintenance and keeping sidewalks clear, levelling sidewalks in some streets, such as the Azaleastraat. There are so many garage driveways that the sidewalk is almost impassable due to sloping everywhere.” “Keep sidewalks clear for pedestrians [...] and sidewalks aren’t accessible everywhere, plants nearly up to facades. Walking in the street is downright dangerous.” “Construction of sidewalks without obstacles, well levelled, and cleaning of the city and weeds.”
		Maintenance and provision of sidewalks for people with limited mobility	35 (16.59%)	“Focus on improving sidewalks. They are dangerously bad for the elderly. With walkers, it’s life-threatening.” “The neighbourhood is not currently wheelchair or rollator-friendly. Not a problem now, but will be later on.” “Repair broken sidewalks so fewer elderly people fall.”
Maintenance and provision of cycling paths	60 (12.15%)			“Prioritize safe bike paths over disrupting the habitat of bats and other animals and build the planned bike paths.” “Better and safer cycling infrastructure like cycling streets, wider bike lanes, better signage at construction sites...” “More bike paths and better maintenance of these as well.” “Replace the cobblestones on the bike path between the trees on the Pintelaan towards Oudenaardsesteenweg with a smooth bike path. It can be slippery due to falling leaves.”

Litter, garbage, and illegal dumping	55 (11.13%)			• “In our street, three benches were once placed, but they are not maintained. Underneath the benches, there is mostly cigarettes and weeds. For years, I kept the area around and under the bench in front of my house clean, but my age doesn’t allow it anymore. I regret that the city neglects it [...] I won’t sit on such a dirty bench.” • “People should be encouraged to throw their waste into designated containers and not on the street, as often happens in my neighbourhood.” • “Tackle illegal dumping!” • “Daily cleanup of litter and illegal dumping, maintenance and keeping sidewalks clear.” • “Effective control on illegal dumping. Illegal dumping is a real plague here.”
Maintenance of streets and public spaces	52 (10.53%)			• “Definitely improve the state of the main streets in Sint-Denijs-Westrem.” • “Better maintenance of streets and squares.” • “Just normal things: clean all streets and maintain them.” • “Create well-paved streets and sidewalks. The historical centre is in terrible shape compared to the surrounding area of the centre.” • “Maintain the streets better: potholes and bumps, unevenly laid tiles...” • “Bring public spaces in good condition for vulnerable pedestrians.”
Maintenance and provision of greenery	48 (9.72%)			• “Keep sidewalks free of weeds.” • “More green parks and trees, and still space and convenience for cyclists.” • “Stop with façade gardens. There isn’t enough space left [...] They are usually neglected after a few years.” • “My street is all concrete with no greenery. Bring calmness with green zones and benches.”
Benches	26 (5.26%)			• “More green spaces and accessible benches for wheelchair users as well, and picnic tables.” • “Walking spaces with plenty of benches.” • “Sheltered bus stops everywhere, and benches here and there along the way.”
Public street lighting	25 (5.06%)			• “Better infrastructure around the station [...], and better lighting. There is little space for pedestrians and less visibility, especially in the rain.” • “Good lighting in the streets in the evening.”

Bike parking in public areas	15 (3.04%)			• “More bike racks and possibly covered ones. Many people in apartments can’t store their bikes indoors.” • “Secure closed bike storage in parking garages and not in places where they obstruct pedestrians or where a small garden area would be better.” • “Bike racks should not be on the street but in an unused space in the park next to the school. This is safer for children and frees up space for parking.” • “There is an urgent need for more bike racks, so bikes don’t have to lean against facades, sometimes two or three stacked over each other.”
Public toilets	2 (0.40%)			• “More rest benches and more public toilets.”

B.9 Themes and quotes – Neighbourhoods and social dynamics

Table B.6: Neighbourhoods and social dynamics (N = 294 – 22.31%)

Theme	Number of quotes	Example quotes
Liveability of the neighbourhood	84 (28.57%)	• "Make the neighbourhood more pleasant." • "More greenery, less noise and less disruptions from businesses." • "More green, small-scale buildings, more quiet zones." • "Make people feel safer and keep drugs out of the neighbourhood." • "More green, better traffic regulations, too many businesses in the area." • "Shielding of R4 (ring road Ghent) with a green buffer and panels against harmful gases and noise."
Parking spaces	58 (19.73%)	• "The introduction of paid parking has caused parking problems in our neighbourhood." • "Create parking spaces so cars don't stay parked in front of our door for 7 to 14 days." • "Stop removing parking spaces." • "Ensure that everyone can park in front of their own door. At least introduce resident parking permits so cars and vans don't park here for days, making our neighbourhood more dangerous." • "Sufficient parking, as I need my car for medical reasons." • "Add more parking spaces and adjust parking meters to allow cash payments." • "Parking spaces so I can get more visits from my family and friends, who are also getting older now." • "More and cheaper parking for visitors. People stay away because of parking costs, which contributes to loneliness."
Social activities and interactions	32 (10.88%)	• "Within the apartment block, I would like to have more activities with other residents." • "Make it cozier. Activities for young retirees. Involvement. We live here as if in a forgotten corner." • "Create small squares as meeting places." • "Organise more neighbourly contact between different age groups." • "A community centre with a social restaurant and café would be wonderful. There is currently nothing like that here."
Participation and attention to older adults	31 (10.54%)	• "More activities for seniors and more input from seniors." • "Address the feeling of insecurity but I don't have a say, so..." • "Since the group of people over 65 is constantly growing, more consideration should be given to them and certainly not in a patronising way. Just because someone is over 65 doesn't mean they have lost all their skills and experiences. Use our age group instead of sidelining it, as often happens..." • "I have no say; everything is decided for me." • "After 42 years of living in Nieuw-Gent, I think I can speak for the neighbourhood." • "This is a rhetorical question. I don't have a say in it, and that will never change?"

Housing conditions	24 (8.16%)	• "I feel good in my home for now, and I realise I am lucky that if I face difficulties, I might look for another home closer to my children or cohabit with someone since I realise that this might be better as I grow older and need more help." • "More cohousing options without tax penalties." • "Affordable housing in the centre of Ghent." • "More regular housing instead of student housing." • "Better and higher quality housing." • "Overall, I enjoy living in Brugse Poort. Just a pity that I only have a small room of 18 square meters, and my housing issue remains unresolved. The private market is too expensive, and I may have to wait another seven years for social housing."
Demographic diversity	18 (6.12%)	• "The Muinkpark is a student neighbourhood with too many student houses and too little parking for permanent residents." • "Better distribute the new Belgians from different backgrounds across the provinces instead of putting them all in one neighbourhood." • "In our street, there are two colleges, a secondary school, a CLB (centre for student support), and a large student housing. As a result, there are hardly any permanent residents with whom to build a neighbour bond. During vacations, all your neighbours go home. If it were up to me, I would create a much healthier and balanced mix of residents instead of large student complexes in one street. We have lived here for 16 months and still don't really know any neighbour." • "Establish positive interaction rules in a very diverse neighbourhood. This applies to everyone, including students, elderly people, and new Belgians. Sometimes we miss basic rules for living together correctly." • "Perhaps limit student housing so that more families can live here." • "It is a shame that I have to take the car to mail a letter." • "More shops in the area." • "Improve access to shops and other facilities within walking distance."
Proximity to services	15 (5.10%)	
Changes in neighbourhood	11 (3.74%)	• "It is painful to see Ghent becoming a city of tourist businesses, chocolatiers, and awful souvenir shops, following in Bruges footsteps." • "I have always been satisfied, and I have liked living in the neighbourhood, but now too many changes are being made that, in my opinion, are not always thought through and are unnecessary. Too much attention is given to a certain target group. You should have left it as it was."
Space constraints and conflicts	10 (3.40%)	• "Provide parking spaces for residents, as placing trees in the street is not safe." • "Return the forests back to agriculture. It is a disgrace that so much beautiful farmland is disappearing in Drongen." • "Newly built streets are too narrow; mirrors are constantly knocked off." • "Parking has also been reduced by charging stations and bike racks where no one ever parks a bike."

Sense of community	6 (2.04%)	• "General policy should focus more on maintaining and enhancing civic mindedness and its glorification. It should encourage a sense of unity rather than "me first" attitudes." • "More community spirit. Follow the rules and don't think you own the place, top level egocentrism." • "More citizens budgets, allowing residents to improve the neighbourhood."
Respect and tolerance	3 (1.02%)	• "More respect and preservation of nature which is often sacrificed for enterprises with commercial motives." • "I think there are no solutions left. People are quick to take to the streets, handing out flyers and organizing protests. Organising or suggesting something results in attacks and violence."
Shared services	2 (0.68)	• "Carsharing in my street." • "More shared cars."

C. Appendix to Chapter 4

C.1 Census population and sample characteristics

Table C.1: Census population and sample characteristics

Characteristic	Census population	Study sample
Gender: women	56.9%	66.5%
Gender: men	43.1%	33.5%
Age: 65-72	41.0%	57.4%
Age: 73-95	59.0%	42.6%
Cluster 1	28.3%	29.4%
Cluster 2	30.2%	29.2%
Cluster 3	25.7%	29.2%
Cluster 4	14.9%	10.9%
Cluster 5	0.9%	1.4%

Note: Population data obtained from Statbel (2023) and Provinciecijfers (2023)

C.2 Indicators of latent variables

Table C.2: Indicators latent variables

BE indicator		SE indicator		Well-being indicator
High-quality walking infrastructure	Walkable with walker or wheelchair	Neighbourhood social cohesion	I know my neighbours well	EQ-5D-index
	Maintained sidewalks		Friendships and relationships in neighbourhood mean a lot	VAS
	Little obstacles on sidewalks		Chat regularly with neighbours	Activity rate compared to peers
	Easy to cross streets		Regular visiting of neighbours	
Walking-friendly environment	Walking-friendly neighbourhood		Neighbours are willing to help each other	
	Enough benches		I can go to neighbours if I need advice	
	Enough greenery		Neighbours would help in emergency	
	Public spaces are pleasant meeting places		I often borrow things from neighbours	
High-quality cycling infrastructure	High-quality cycling lanes		My neighbours are trustworthy	
	Maintained cycling infrastructure		My neighbours get along	
	Sufficient cycling infrastructure		My neighbourhood is close-knit	
	Maintained streets and squares		Living in this neighbourhood gives a sense of community	
Cycling-friendly environment	Cycling-friendly environment	Attractive living environment	I live in an attractive neighbourhood	
	Enjoyable cycling		I feel like I belong to this neighbourhood	
	Feeling safe in traffic as a cyclist		I would like to move if given the opportunity	
Busy traffic	Busy traffic causing unsafe cycling experiences		Planning to live here for several years	
	A lot of traffic		Good place to bring up children	
	A lot of freight traffic	Encouragement for physical activity	I can count on people to be physically active with me	

	Nuisance due to traffic		Family and friends encourage me to be active	
Traffic safety	Few traffic accidents		People important to me encourage me to participate in physical activities	
	Enough public lighting		Family and friends participate in physical activities regularly	
	Feeling safe in traffic as a pedestrian		Seeing others participating in physical activities makes me want to participate	
	Feeling safe in traffic as a car user	Satisfaction with friendships in the neighbourhood	Satisfied with the number of friends in the neighbourhood	
	Feeling safe on public transport		Visit friends at their place	
Positive parking experiences	Accessible by car		Satisfied with the number of people I know in the neighbourhood	
	Easily finding parking spots for residents	Satisfaction with events in the neighbourhood	Older adults are informed about activities	
	Easily finding parking spots for visitors		Sufficient activities are aimed at or adapted to older adults	
High-quality public transport	Satisfied with public transport		Activities are easily accessible	
	Enough public transport service		Older adults are sufficiently involved in the organisation of activities	
	Sufficiently frequent public transport			
	Well-equipped public transport stops			
	Accessible public transport stops			
	Enough seats on public transport			
	Easy to find information on public transport for older people			

C.3 Multigroup SEM walking model

Table C.3: Multigroup SEM age

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	12348.708	4234	0.911	0.059	0.073
Metric Invariance	12883.700	4297	0.906	0.061	0.074
Scalar Invariance	12869.433	4349	0.906	0.060	0.074
Strict Invariance	12924.850	4416	0.906	0.060	0.074

Note: Group 1: range 65-72 years old; Group 2: range 73-95 years old

Table C.4: Multigroup SEM gender

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	12393.712	4234	0.910	0.060	0.073
Metric Invariance	12883.707	4297	0.906	0.061	0.074
Scalar Invariance	12867.422	4349	0.906	0.060	0.074
Strict Invariance	12883.524	4416	0.907	0.059	0.074

Note: Group 1: female; Group 2: male

Table C.5: Multigroup SEM household size

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	12375.047	4234	0.910	0.059	0.073
Metric Invariance	12859.792	4297	0.906	0.061	0.074
Scalar Invariance	12861.705	4349	0.906	0.060	0.074
Strict Invariance	12870.064	4416	0.907	0.059	0.074

Note: Group 1: single-person household; Group 2: multi-person household

Table C.6: Multigroup SEM income

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	10511.276	4234	0.917	0.057	0.073
Metric Invariance	10897.534	4297	0.913	0.058	0.074
Scalar Invariance	10902.699	4349	0.913	0.058	0.074
Strict Invariance	10909.253	4416	0.914	0.057	0.074

Note: Group 1: lower-income households; Group 2: higher-income households

Table C.7: Multigroup SEM main travel mode

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	16309.653	8468	0.911	0.058	0.082
Metric Invariance	17923.235	8657	0.895	0.063	0.086
Scalar Invariance	18000.078	8813	0.896	0.062	0.087
Strict Invariance	18072.014	9014	0.898	0.061	0.087

Note: Group 1: on foot; Group 2: bike, Group 3: car; Group 4: public transport

Table C.8: Multigroup SEM cluster of living neighbourhood

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10657.759	2117	0.906	0.061	0.068
Configural model	15156.337	8468	0.918	0.056	0.083
Metric Invariance	16554.946	8657	0.903	0.061	0.086
Scalar Invariance	16588.563	8813	0.905	0.060	0.087
Strict Invariance	16633.882	9014	0.907	0.058	0.087

Note: Group 1: city centre; Group 2: neighbourhoods adjacent to city centre; Group 3: suburban residential neighbourhoods (N); Group 4: suburban residential neighbourhoods (S)

C.4 Multigroup SEM cycling model

Table C.9: Multigroup SEM age

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	10407.324	4232	0.915	0.052	0.071
Metric Invariance	10866.846	4295	0.910	0.053	0.073
Scalar Invariance	10866.846	4346	0.911	0.053	0.073
Strict Invariance	10882.724	4416	0.911	0.052	0.073

Note: Group 1: range 65-72 years old; Group 2: range 73-95 years old

Table C.10: Multigroup SEM gender

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	10383.139	4232	0.916	0.052	0.072
Metric Invariance	10788.759	4295	0.911	0.053	0.073
Scalar Invariance	10788.759	4346	0.911	0.052	0.073
Strict Invariance	10803.027	4413	0.912	0.052	0.073

Note: Group 1: female; Group 2: male

Table C.11: Multigroup SEM household size

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	10361.098	4232	0.915	0.052	0.071
Metric Invariance	10787.286	4295	0.910	0.053	0.073
Scalar Invariance	10787.286	4346	0.911	0.052	0.073
Strict Invariance	10795.062	4413	0.912	0.052	0.073

Note: Group 1: single-person household; Group 2: multi-person household

Table C.12: Multigroup SEM income

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	8888.742	4232	0.924	0.049	0.072
Metric Invariance	9205.960	4295	0.920	0.050	0.073
Scalar Invariance	9205.960	4346	0.921	0.050	0.073
Strict Invariance	9214.600	4413	0.922	0.049	0.073

Note: Group 1: lower-income households; Group 2: higher-income households

Table C.13: Multigroup SEM main travel mode

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	13312.042	8464	0.932	0.046	0.081
Metric Invariance	15147.413	8653	0.909	0.053	0.085
Scalar Invariance	15147.413	8806	0.911	0.052	0.085
Strict Invariance	15184.032	9007	0.914	0.050	0.085

Note: Group 1: on foot; Group 2: bike, Group 3: car; Group 4: public transport

Table C.14: Multigroup SEM cluster of living neighbourhood

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10764.461	2117	0.906	0.061	0.068
Configural model	12716.142	8464	0.936	0.045	0.081
Metric Invariance	14027.382	8653	0.919	0.050	0.085
Scalar Invariance	14027.382	8806	0.921	0.049	0.085
Strict Invariance	14073.909	9007	0.923	0.048	0.085

Note: Group 1: city centre; Group 2: neighbourhoods adjacent to city centre; Group 3: suburban residential neighbourhoods (N); Group 4: suburban residential neighbourhoods (S)

C.5 Multigroup SEM car use model

Table C.15: Multigroup SEM age

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	12380.781	4230	0.907	0.060	0.060
Metric Invariance	12915.172	4293	0.905	0.061	0.074
Scalar Invariance	12917.866	4345	0.906	0.060	0.074
Strict Invariance	12949.691	4412	0.906	0.060	0.074

Note: Group 1: range 65-72 years old; Group 2: range 73-95 years old

Table C.16: Multigroup SEM gender

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	12473.759	4230	0.910	0.060	0.073
Metric Invariance	12948.368	4293	0.905	0.061	0.074
Scalar Invariance	12965.502	4345	0.905	0.060	0.074
Strict Invariance	12987.819	4412	0.906	0.060	0.074

Note: Group 1: female; Group 2: male

Table C.17: Multigroup SEM household size

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	12449.016	4230	0.909	0.060	0.073
Metric Invariance	12935.870	4293	0.905	0.061	0.074
Scalar Invariance	12976.216	4345	0.905	0.060	0.075
Strict Invariance	12999.372	4412	0.906	0.060	0.075

Note: Group 1: single-person household; Group 2: multi-person household

Table C.18: Multigroup SEM income

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	10568.305	4230	0.916	0.057	0.073
Metric Invariance	10948.928	4293	0.912	0.058	0.074
Scalar Invariance					
Strict Invariance					

Note: Group 1: lower-income households; Group 2: higher-income households
 Scalar and strict invariance models could not be estimated for household income groups due to insufficient observations and a high share of missing ('unknown') responses.

Table C.19: Multigroup SEM main travel mode

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	12736.587	4230	0.906	0.061	0.073
Metric Invariance	13270.399	4293	0.900	0.062	0.075
Scalar Invariance	13310.511	4345	0.900	0.062	0.075
Strict Invariance	13335.900	4412	0.901	0.061	0.075

Note: Group 1: active modes; Group 2: motorised modes

Table C.20: Multigroup SEM cluster of living neighbourhood

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10706.960	2115	0.906	0.061	0.068
Configural model	13391.794	6345	0.915	0.058	0.078
Metric Invariance	14546.732	6471	0.902	0.061	0.081
Scalar Invariance	14590.995	6575	0.903	0.061	0.081
Strict Invariance	14621.776	6709	0.904	0.060	0.081

Note: Group 1: city centre; Group 2: neighbourhoods adjacent to city centre; Group 3: suburban residential neighbourhoods

C.6 Multigroup SEM public transport use model

Table C.21: Multigroup SEM age

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	12457.153	4232	0.910	0.060	0.073
Metric Invariance	12991.774	4295	0.905	0.061	0.074
Scalar Invariance	13021.491	4347	0.905	0.061	0.074
Strict Invariance	13048.570	4414	0.905	0.060	0.074

Note: Group 1: range 65-72 years old; Group 2: range 73-95 years old

Table C.22: Multigroup SEM gender

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	12506.869	4232	0.909	0.060	0.073
Metric Invariance	12978.519	4295	0.905	0.061	0.074
Scalar Invariance	12990.753	4347	0.905	0.061	0.074
Strict Invariance	13004.712	4414	0.906	0.060	0.074

Note: Group 1: female; Group 2: male

Table C.23: Multigroup SEM household size

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	12473.064	4232	0.909	0.060	0.073
Metric Invariance	12947.849	4295	0.905	0.061	0.074
Scalar Invariance	12993.465	4347	0.905	0.060	0.074
Strict Invariance	13008.570	4414	0.906	0.060	0.074

Note: Group 1: single-person household; Group 2: multi-person household

Table C.24: Multigroup SEM income

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	10599.344	4232	0.916	0.058	0.073
Metric Invariance	10982.088	4295	0.911	0.059	0.075
Scalar Invariance	10987.200	4347	0.912	0.058	0.075
Strict Invariance	11000.584	4414	0.913	0.057	0.075

Note: Group 1: lower-income households; Group 2: higher-income households

Table C.25: Multigroup SEM main travel mode

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	16322.202	8464	0.911	0.058	0.082
Metric Invariance	17939.234	8653	0.895	0.063	0.086
Scalar Invariance	18122.551	8809	0.895	0.062	0.087
Strict Invariance	18184.846	9010	0.897	0.061	0.087

Note: Group 1: on foot; Group 2: bike, Group 3: car; Group 4: public transport

Table C.26: Multigroup SEM cluster of living neighbourhood

Model	X2	Df	CFI	RMSEA	SRMR
Overall model	10765.527	2116	0.905	0.061	0.068
Configural model	13495.788	6348	0.913	0.058	0.078
Metric Invariance	14656.293	6474	0.901	0.062	0.082
Scalar Invariance	14721.477	6578	0.902	0.061	0.082
Strict Invariance	14753.028	4414	0.903	0.060	0.082

Note: Group 1: city centre; Group 2: neighbourhoods adjacent to city centre; Group 3: suburban residential neighbourhoods

D. Appendix to Chapter 5

D.1 Geocoding process

Address records were first pre-processed to normalise house numbers (removing box and apartment suffixes and replacing placeholder tokens such as "no house number") and to strip additions from street name fields. Unique addresses were extracted and processed once, with results including failures stored in a persistent cache and propagated to all duplicate records across years, avoiding redundant processing.

In the first phase, each address was submitted to the Flemish government's Geopunt geocoding API (<https://geo.api.vlaanderen.be>), which covers the full Flemish address register. Addresses not resolved by Geopunt were passed to the OpenStreetMap Overpass API (<https://overpass-api.de>) and queried within a municipality-scoped bounding area to retrieve named street or point-of-interest centroids. Addresses for which no valid house number was recorded were handled by a dedicated street-centre fallback, which retrieved the geometric centroid of the matched street segment via Overpass or, where that failed, via Geopunt without a house number component.

Addresses remaining unresolved after all direct queries entered a second phase. The core challenge was that a single address field frequently encoded not a street name but a facility name (a nursing home, hospital, or medical practice), sometimes combined with a street name in the same string. This type of ambiguity, where place names, institutional identifiers, and street names are mixed in free-text fields, naturally lends itself well to language model-based text understanding (Cruz et al., 2021), and the use of transformer-based language models to improve geocoding outcomes for messy or irregular address inputs has been demonstrated in applied settings (Berragan et al., 2022; Maheshwari et al., 2025). A transformer-based large language model (claude-3-haiku-20240307, Anthropic) was therefore prompted to decompose each ambiguous entry into two candidate sets: cleaned street name variants for re-submission to Geopunt, and plausible facility name variants for re-submission to Overpass. The model was queried via the Anthropic API; only the address string and municipality name were transmitted, with no personal or record-level data included. The temperature parameter was set to 0.3 and the maximum output length to 200 tokens. The model was instructed to return a structured JSON object with separate address and OSM keys, and the prompt was structured as a few-shot sequence presenting known input-output pairs to ground the model's output in local Flemish naming conventions. Each example demonstrated how an ambiguous field should be split: an entry combining a care facility name with a street name was shown to yield the bare street name as the address variant, and the facility name in its common abbreviated, full, and alternative forms as the OSM variants. Up to two address variants and four OSM variants were retained per entry. Generated variations were cached so that identical ambiguous strings encountered across years were not reprocessed.

From a data-protection perspective, the information transmitted through the Anthropic API was minimised to what was necessary for the geocoding task. Only address strings and municipality names were transmitted; no names, participant identifiers, trip characteristics, or other attributes from the Mobitwin

dataset were included in the API requests. According to Anthropic's data-handling policy applicable to its commercial API, API inputs and outputs were not used for model training by default and were automatically deleted from Anthropic's backend within 30 days, except where longer retention was required for usage-policy enforcement or legal obligations.

D.2 Destination categories

Table D.1: Destination categories

Goal	Source	Year	Destinations (N)
Hospital	Digitaal Vlaanderen	2025	196
Adult day care centre	Digitaal Vlaanderen	2025	319
Socio-cultural activities	OpenStreetMap	2025	1225
Groceries	Digitaal Vlaanderen	2015	40000
Paramedical: speech therapists	De Sociale Kaart (Departement Zorg)	2025	1736
Paramedical: physiotherapists	De Sociale Kaart (Departement Zorg)	2025	4859
Paramedical: dieticians	De Sociale Kaart (Departement Zorg)	2025	1545
Paramedical: occupational therapist	De Sociale Kaart (Departement Zorg)	2025	227
General practitioner	De Sociale Kaart (Departement Zorg)	2025	3185
Hairdresser	OpenStreetMap	2025	2215
Restaurant	OpenStreetMap	2025	7117
Dentist	De Sociale Kaart (Departement Zorg)	2025	455
Administrative trip	OpenStreetMap	2025	536
Local service centre	Digitaal Vlaanderen	2025	345
School	Digitaal Vlaanderen	2025	9400
Special needs education	Digitaal Vlaanderen	2025	796
Funeral homes (crematoriums)	OpenStreetMap	2025	16
Public transport stop	Digitaal Vlaanderen	2025	29714
Nursing home	Digitaal Vlaanderen	2025	813
Pharmacies	De Sociale Kaart (Departement Zorg)	2025	2491
Cemeteries	Onroerend erfgoed Vlaanderen	2025	720

D.3 Rural-Urban Flemish classification

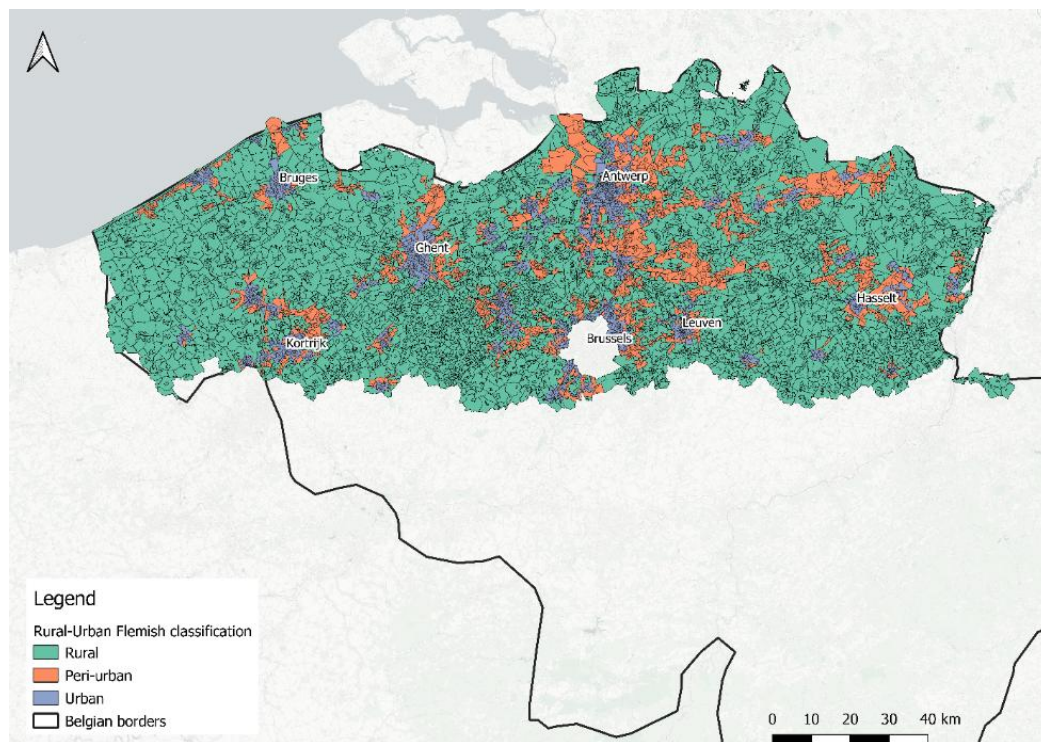


Figure D.1: Rural-Urban Flemish classification

D.4 Chi-square tests of independence

Table D.2: Chi-square test adult day care centres

	Access	No access
Rural	6632	11252
Suburban	2936	5997
Urban	8616	4829
Chi ² test	2960.1	
Df	2	
P value	< 2.2e-16	
Cramer's V	0.271	

Table D.3: Chi-square test hospitals

	Access	No access
Rural	1751	16133
Suburban	1661	7272
Urban	6029	7416
Chi ² test	5403.7	
Df	2	
P value	< 2.2e-16	
Cramer's V	0.366	

Table D.4: Chi-square test socio-cultural activities

	Access	No access
Rural	9108	8776
Suburban	5211	3722
Urban	11415	2030
Chi ² test	3996	
Df	2	
P value	< 2.2e-16	
Cramer's V	0.315	

Table D.5: Chi-square test grocery stores

	Access	No access
Rural	16525	1359
Suburban	8645	288
Urban	13406	39
Chi ² test	1048.4	
Df	2	
P value	< 2.2e-16	
Cramer's V	0.161	

D.5 Comparison cumulative accessibility results 2023 and 2024

Table D.6: Comparison cumulative accessibility results 2023 and 2024

Destination	Zero-access 2023	Zero-access 2023 (%)	Zero-access 2024	Zero-access 2024 (%)	Difference % 2024- 2023
Adult day care centres	23233	57.66	22104	54.86	-2.80
Cemeteries	26061	64.68	26187	65.00	0.31
Crematoriums	39913	99.06	39941	99.13	0.07
Dentists	27254	67.64	27351	67.89	0.24
Dieticians	11085	27.51	11919	29.58	2.07
General practitioners	5414	13.44	6152	15.27	1.83
Groceries	1525	3.79	1698	4.21	0.43
Hairdressers	11369	28.22	11993	29.77	1.55
Hospitals	30567	75.87	30849	76.57	0.70
Local service centres	22791	56.57	22833	56.67	0.10
Nursing homes	14662	36.39	14779	36.68	0.29
Occupational therapists	29576	73.41	29498	73.21	-0.19
Pharmacies	5699	14.14	6201	15.39	1.25
Physiotherapists	4116	10.22	4543	11.28	1.06
Public transport stops	1112	2.76	1149	2.85	0.09
Restaurants	5895	14.63	6922	17.18	2.55
Schools	3761	9.33	4264	10.58	1.25
Socio-cultural activities	13574	33.69	14548	36.11	2.42
Special needs education	24738	61.40	24706	61.32	-0.08
Speech therapists	10323	25.62	10788	26.78	1.15
Town halls	15539	38.57	15555	38.61	0.04

D.6 Sensitivity analysis

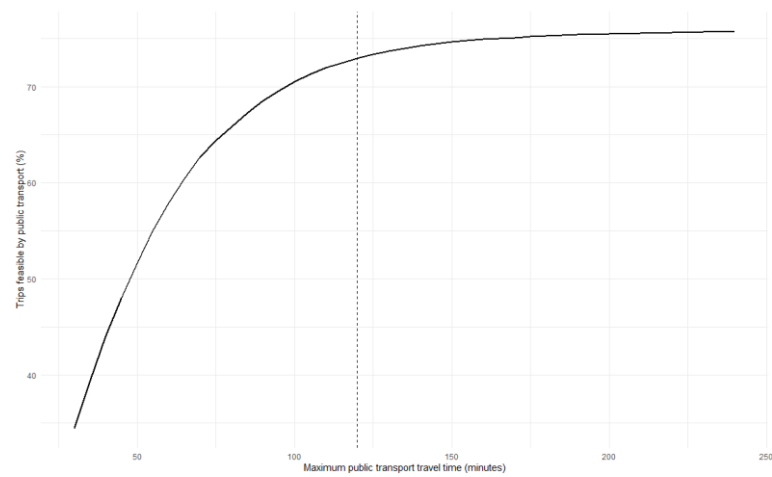


Figure D.2: Sensitivity of public transport trip feasibility to the maximum travel time threshold

Note: The dashed vertical line indicates the 120-minute maximum travel time threshold applied in the main analysis.

D.7 Binary logistic regression results

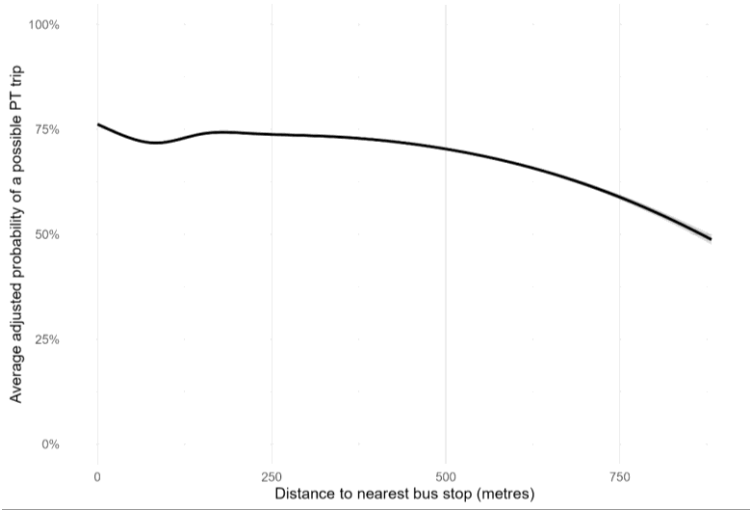


Figure D.3: Average adjusted association with distance to nearest bus stop

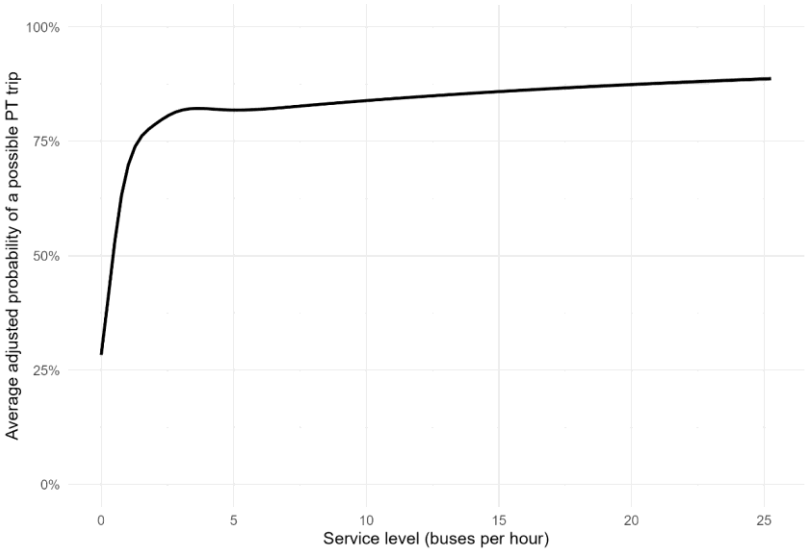


Figure D.4: Average adjusted association with service level of nearest bus stops

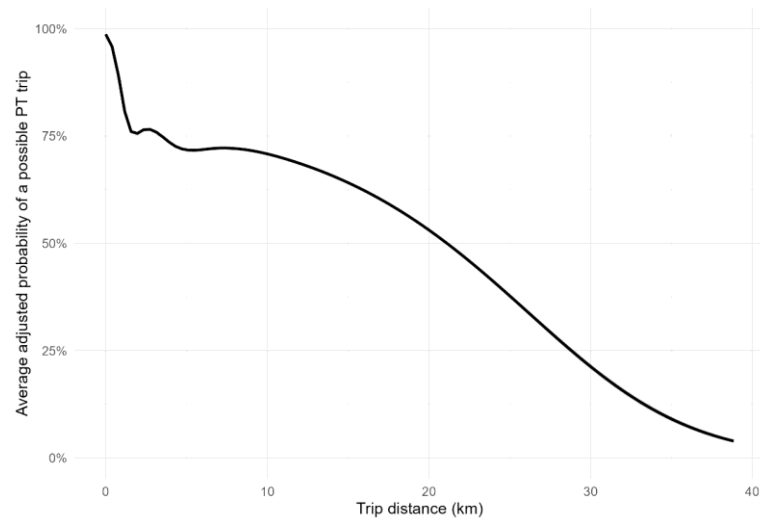


Figure D.5: Average adjusted association with trip distance